

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 1: Noticing the Phenomenon & Building the Need: What Tool Is Missing?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	B	C	A	C	D	D

USE AFTER: PREDICT PHASE — Predict: groups' 'best attempt' using only algebra and Pythagoras on the mini-whiteboard

Q1. A student stands 40 m from the base of a straight transmission tower and estimates that the angle up to the top is about 50° . The class tries to find the height using only algebra and Pythagoras. Which statement best explains why they get stuck?

- A. Pythagoras needs two sides of a right-angled triangle, but here only one side (40 m) and an angle are known.**
- B. Pythagoras can be used directly because 40 m and 50° are both known.
- C. The tower does not form a right-angled triangle, so no method could ever work.
- D. Pythagoras only works for triangles taller than 40 m.

Correct answer: A. Pythagoras links sides to sides, so it needs two known sides. Only one side and one angle are given, so students need a new tool that links an angle to side lengths.

USE AFTER: OBSERVE PHASE — Part A: Environment Scan (Triangle Hunt, labelling O, A and H)

Q2. A wheelchair ramp forms a right-angled triangle with the level ground and a vertical wall. The ramp is the sloping side. You stand at the angle where the ramp meets the ground. Which side is the ADJACENT side to that angle?

- A. The ramp (the sloping side)
- B. The level ground**
- C. The right angle itself
- D. The vertical wall

Correct answer: B. The adjacent side lies next to the marked angle and is not the hypotenuse. Here it is the ground. The ramp is the hypotenuse and the wall is the opposite side.

USE AFTER: OBSERVE PHASE — Part B: Ratio Discovery Activity (computing $O \div H$, $A \div H$, $O \div A$)

Q3. In Part B, a group draws a right-angled triangle with a 40° angle. They measure the opposite side as 3.9 cm, the adjacent side as 4.6 cm and the hypotenuse as 6.0 cm. What is $O \div H$ to two decimal places?

- A. 0.77
- B. 0.85
- C. 0.65**
- D. 1.54

Correct answer: C. $O \div H = 3.9 \div 6.0 = 0.65$. The value 0.77 is $A \div H$, 0.85 is $O \div A$, and 1.54 is $H \div O$ (the ratio upside down).

USE AFTER: OBSERVE PHASE — Part B: Ratio Discovery Activity (group table and 'circle any pattern')

Q4. A group draws three right-angled triangles that all have a 40° angle. Their hypotenuses are about 6 cm, 9 cm and 12 cm. They find $O \div H = 0.64, 0.65$ and 0.64 . What does this result show?

- A. The $O \div H$ values are almost the same even though the triangles are different sizes.**
- B. $O \div H$ is always equal to the length of the hypotenuse.
- C. $O \div H$ gets bigger as the triangle gets bigger.
- D. The group must have measured wrongly, because different triangles should give very different values.

Correct answer: A. The three values differ only by a tiny amount, which is measurement error. The data show the ratio stays nearly constant for the same 40° angle, although the triangle sizes differ.

USE AFTER: EXPLAIN PHASE — Explain: returning to the mast problem and writing 'height $\div$ 40 = $\tan 50^\circ$ '

Q5. In the mast problem, the right angle is at the base of the mast. The observer stands 40 m from the base, and the angle from the observer up to the top is 50° . The height is the side opposite the 50° angle, and the 40 m is the side adjacent to it. Which equation connects the height to the 40 m?

- A. height $\div$ 40 = $\sin 50^\circ$
- B. height $\div$ 40 = $\cos 50^\circ$
- C. height $\div$ 40 = $\tan 50^\circ$**
- D. height $\div$ 40 = $\tan 40^\circ$

Correct answer: C. Opposite $\div$ adjacent is the tangent ratio, so height $\div$ 40 = $\tan 50^\circ$. Sine uses the hypotenuse and cosine uses the adjacent over the hypotenuse, and neither fits here.

USE AFTER: DQB PHASE — DQB: writing one question about the mast problem or the ratios

Q6. Which of these is the best question to post on the Driving Question Board for the mast problem?

- A. Is the mast in Nakuru?
- B. Why do we have to do trigonometry?
- C. What is maths?
- D. How can we find the value of $\tan 50^\circ$ without measuring a huge triangle?**

Correct answer: D. A good question is specific and linked to the phenomenon. It names what is still missing (the value of the $\tan 50^\circ$ ratio) and points to the next steps in the unit. The other questions are vague or not investigable.

USE AFTER: MODEL BUILDING PHASE — Model: initial annotated pencil sketch of the mast scenario

Q7. When drawing a model of the mast scenario (a vertical mast standing on level ground, with the observer 40 m away), where should the right angle be marked?

- A. Halfway along the 40 m line
- B. At the top of the mast
- C. At the observer's feet, where the 50° angle is
- D. At the base of the mast, where the vertical mast meets the level ground**

Correct answer: D. The mast stands straight up from the level ground, so the right angle is at its base. The 50° angle is at the observer's position.